

FOSD — Unit 2

CPU Scheduling

GATE CSE PYQ source-page collection • 2017–2026

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- The selected question numbers are listed below; neighboring questions may remain visible because the original page is preserved.
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Included source pages

CS2019 — source page 13 — selected Q41
CS2020 — source page 15 — selected Q50
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- Q.39 Consider the following snapshot of a system running n concurrent processes. Process i is holding X_i instances of a resource R , $1 \leq i \leq n$. Assume that all instances of R are currently in use. Further, for all i , process i can place a request for at most Y_i additional instances of R while holding the X_i instances it already has. Of the n processes, there are exactly two processes p and q such that $Y_p = Y_q = 0$. Which one of the following conditions guarantees that no other process apart from p and q can complete execution?

- (A) $X_p + X_q < \text{Min} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
 (B) $X_p + X_q < \text{Max} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
 (C) $\text{Min}(X_p, X_q) \geq \text{Min} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
 (D) $\text{Min}(X_p, X_q) \leq \text{Max} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$

- Q.40 Consider the following statements:

- I. The smallest element in a max-heap is always at a leaf node
 II. The second largest element in a max-heap is always a child of the root node
 III. A max-heap can be constructed from a binary search tree in $\Theta(n)$ time
 IV. A binary search tree can be constructed from a max-heap in $\Theta(n)$ time

Which of the above statements are TRUE?

- (A) I, II and III (B) I, II and IV (C) I, III and IV (D) II, III and IV

- Q.41 Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P1	P2	P3	P4
Arrival time	0	1	3	4
CPU burst time	3	1	3	Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is _____.

- Q.42 The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointers. The disk block size is 4 kB, and the disk block address is 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) _____ GB.

Consider the following C functions.

```
int tob(int b, int* arr){
    int i;
    for(i=0; b>0; i++){
        if(b%2) arr[i]=1;
        else    arr[i]=0;
        b = b/2;
    }
    return(i);
}

int pp(int a,int b) {
    int arr[20];
    int i, tot = 1, ex, len;
    ex = a;
    len = tob(b,arr);
    for(i=0; i<len; i++){
        if(arr[i]==1)
            tot = tot * ex;
        ex = ex * ex;
    }
    return(tot);
}
```

The value returned by `pp(3, 4)` is _____.

Q.No. 49 Consider a graph $G = (V, E)$, where $V = \{v_1, v_2, \dots, v_{100}\}$, $E = \{(v_i, v_j) | 1 \leq i < j \leq 100\}$, and weight of the edge (v_i, v_j) is $|i - j|$. The weight of minimum spanning tree of G is _____.

Q.No. 50 Consider the following set of processes, assumed to have arrived at time 0. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order P_1, P_2, P_3, P_4 .

Processes	P_1	P_2	P_3	P_4
Burst time (in ms)	8	7	2	4

If the time quantum for RR is 4 ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to 2 decimal places) is _____.

Q.No. 51 Consider the following language.

$$L = \{x \in \{a, b\}^* \mid \text{number of } a\text{'s in } x \text{ is divisible by 2 but not divisible by 3}\}$$

The minimum number of states in a DFA that accepts L is _____.

Q.No. 52 Graph G is obtained by adding vertex s to $K_{3,4}$ and making s adjacent to every vertex of $K_{3,4}$. The minimum number of colours required to edge-colour G is _____.

Q.No. 53



Computer Science and Information Technology (CS, Set-1)

Q.24	Consider the following representation of a number in IEEE 754 single-precision floating point format with a bias of 127. $S : 1 \quad E : 10000001 \quad F : 11110000000000000000000$ Here S, E and F denote the sign, exponent and fraction components of the floating point representation. The decimal value corresponding to the above representation (rounded to 2 decimal places) is _____.
Q.25	Three processes arrive at time zero with CPU bursts of 16, 20 and 10 milliseconds. If the scheduler has prior knowledge about the length of the CPU bursts, the minimum achievable average waiting time for these three processes in a non-preemptive scheduler (rounded to nearest integer) is _____ milliseconds.

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Q.41	Consider three floating point numbers A, B and C stored in registers R_A, R_B and R_C, respectively as per IEEE-754 single precision floating point format. The 32-bit content stored in these registers (in hexadecimal form) are as follows. <table><tr><td>$R_A = 0xC1400000$</td><td>$R_B = 0x42100000$</td><td>$R_C = 0x41400000$</td></tr></table> Which one of the following is FALSE?	$R_A = 0xC1400000$	$R_B = 0x42100000$	$R_C = 0x41400000$
$R_A = 0xC1400000$	$R_B = 0x42100000$	$R_C = 0x41400000$		
(A)	$A + C = 0$			
(B)	$C = A + B$			
(C)	$B = 3C$			
(D)	$(B - C) > 0$			
Q.42	Consider four processes P, Q, R, and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?			
(A)	$P = 4, Q = 10, R = 6, S = 2$			
(B)	$P = 2, Q = 9, R = 5, S = 1$			
(C)	$P = 4, Q = 12, R = 5, S = 4$			
(D)	$P = 3, Q = 7, R = 7, S = 3$			

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Q.27	Which one or more of the following CPU scheduling algorithms can potentially cause starvation?
(A)	First-in First-Out
(B)	Round Robin
(C)	Priority Scheduling
(D)	Shortest Job First

- Q.37 Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time.

A (0, 10), B (2, 6), C (4, 3), and D (6, 7).

Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF) CPU scheduling algorithms are applied to the processes?

- (A) SRTF = 6, NP-SJF = 7
- (B) SRTF = 6, NP-SJF = 7.5
- (C) SRTF = 7, NP-SJF = 7.5
- (D) SRTF = 7, NP-SJF = 8.5

- Q.38 Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

- (A) 10.12.2.0/23
- (B) 10.12.2.0/24
- (C) 10.12.0.0/22
- (D) 10.12.2.0/22

Q.38	A computer has two processors, M_1 and M_2. Four processes P_1, P_2, P_3, P_4 with CPU bursts of 20, 16, 25, and 10 milliseconds, respectively, arrive at the same time and these are the only processes in the system. The scheduler uses non-preemptive priority scheduling, with priorities decided as follows: • M_1 uses priority of execution for the processes as, $P_1 > P_3 > P_2 > P_4$, i.e., P_1 and P_4 have highest and lowest priorities, respectively. • M_2 uses priority of execution for the processes as, $P_2 > P_3 > P_4 > P_1$, i.e., P_2 and P_1 have highest and lowest priorities, respectively. A process P_i is scheduled to a processor M_k, if the processor is free and no other process P_j is waiting with higher priority. At any given point of time, a process can be allocated to any one of the free processors without violating the execution priority rules. Ignore the context switch time. What will be the average waiting time of the processes in milliseconds?
(A)	9.00
(B)	8.75
(C)	6.50
(D)	7.50

Q.25	Let G_1, G_2 be Context Free Grammars (CFGs) and R be a regular expression. For a grammar G, let $L(G)$ denote the language generated by G. Which ONE among the following questions is decidable?
(A)	Is $L(G_1) = L(G_2)$?
(B)	Is $L(G_1) \cap L(G_2) = \emptyset$?
(C)	Is $L(G_1) = L(R)$?
(D)	Is $L(G_1) = \emptyset$?
Q.26	Processes P_1, P_2, P_3, P_4 arrive in that order at times 0, 1, 2, and 8 milliseconds respectively, and have execution times of 10, 13, 6, and 9 milliseconds respectively. Shortest Remaining Time First (SRTF) algorithm is used as the CPU scheduling policy. Ignore context switching times. Which ONE of the following correctly gives the average turnaround time of the four processes in milliseconds?
(A)	22
(B)	15
(C)	37
(D)	19

Q.22	Which one of the following protocols may need to broadcast some of its messages?
(A)	SMTP
(B)	FTP
(C)	DHCP
(D)	HTTP
Q.23	Which one of the following CPU scheduling algorithms cannot be preemptive?
(A)	Shortest Remaining Time First (SRTF) Scheduling
(B)	First Come First Serve (FCFS) Scheduling
(C)	Round Robin Scheduling
(D)	Priority Scheduling

Q.63	Consider the following program snippet. Assume that the program compiles and runs successfully. Further, assume that the <code>fork()</code> system call is always successful in creating a process. <pre> int main () { int i; for (i = 0; i < 3; i++){ if (fork() == 0){ continue; } break; } printf("Hello!"); return 0; } </pre> The total number of times that the <code>printf</code> statement gets executed is _____. (answer in integer)
Q.64	Consider a CPU that has to execute two types of processes. The first type, Actuators (A), requires a CPU burst of 6 seconds. The second type, Controllers (C), requires a CPU burst of 8 seconds. A new process of type A arrives at time $t = 10, 20, 30, 40,$ and 50 (in seconds). Similarly, a new process of type C arrives at time $t = 11, 22, 33, 44,$ and 55 (in seconds). The CPU scheduling policy is First Come First Serve (FCFS). The first process of type A starts running at $t = 10$ seconds. The average waiting time (in seconds) for the 10 processes is _____. (rounded off to one decimal place)
Q.65	Consider a relational database schema with a relation $R(A, B, C, D)$. If $\{A, B\}$ and $\{A, C\}$ are the only two candidate keys of the relation R, then the number of superkeys of relation R is _____. (answer in integer)